Resolution Boundaries as Signals of Operator Stratification

1 Resolution Boundaries as Signals of Operator Stratification

The results reported in this work represent emergent invariants of the UNNS substrate at the resolution of the currently implemented τ-field operators. Residual offsets and incomplete convergence are therefore not interpreted as modeling deficiencies, but as resolution boundaries indicating the presence of additional operator strata.

In the UNNS framework, such boundaries mark locations where further structural detail becomes admissible only through operator extension, not parameter adjustment. The role of the present work is not to eliminate these boundaries, but to identify them, characterize their stability, and use them as signals of substrate depth.

The emergence of stable invariants under τ-field dynamics establishes that the recursive substrate admits projection into persistent quantities that are robust under parameter variation. At the same time, the appearance of non-vanishing residuals that resist further convergence indicates that the current operator layer has reached the limit of its admissible resolution.

This interpretation departs from conventional approaches in which residuals are treated as errors to be minimized. Within UNNS, a stable residual is not necessarily a failure of the model, but may instead encode information about the internal structure of the substrate that lies beyond the reach of the active operators.

The distinction between numerical error and structural boundary is therefore central. Numerical errors typically fluctuate under changes in discretization, integration scheme, or parameter tuning. Structural boundaries, by contrast, remain stable across such variations and persist even as the dynamics converge in all other respects.

The identification of resolution boundaries requires three conditions:

- convergence of the relevant τ-invariants,
- stability of the residual under parameter variation, and
- insensitivity of the residual to numerical refinement.

When these conditions are satisfied, the residual can be interpreted as a signal that the current operator layer has exhausted its descriptive power for the structure under consideration.

The remainder of this work examines such boundaries across multiple regimes, including nuclear and hadronic scales, vacuum structure, and global invariant formation. These boundaries are organized into gap classes, each corresponding to a distinct type of limitation and guiding a different mode of extension of the UNNS framework.

Relation to Emergent Constants. The operator-stratification framework developed here provides a formal interpretive context for the stable residuals reported in *Physical Constants as Emergent Invariants of Recursive Substrate Dynamics*, without modifying the empirical results themselves.

Epistemic Sequencing. The κ-Operator is not postulated a priori. Its introduction is motivated by boundary analysis of stable, resolution-independent residuals observed under τ-field projection. Once parameter variation and numerical refinement failed to reduce these residuals without disrupting established invariants, it became evident that the limitation was structural rather than numerical or projective.

Accordingly, the κ-Operator is introduced as a pre-projective coherence regulator whose necessity is inferred from the observed behavior of the substrate, rather than assumed in advance.

2 Nuclear and Hadronic Scales

Gap Class A: Structural Depth

Application of the τ-field operators to nuclear and hadronic regimes produces stable convergence behavior for the majority of observables examined. However, a persistent residual of approximately 1.82% remains in the convergence of the proton–electron mass ratio μ.

This residual exhibits three critical properties. First, it is stable across variations of numerical resolution and integration parameters. Second, it remains invariant under moderate changes to the initial conditions of the

recursive dynamics. Third, it does not correlate with discretization artifacts or truncation depth.

These properties distinguish the residual from numerical error and identify it as a structural feature of the projection. Within the UNNS framework, this behavior indicates that the active τ-field operator captures large-scale binding structure but does not fully resolve sub-nuclear organization.

In particular, quark-level binding and confinement dynamics appear as internal modes that are not expressible within the present operator layer. The τ-field produces a coherent projection of nuclear-scale structure, but the remaining offset signals that additional internal constraints remain unaccounted for.

The persistence of this residual under parameter variation indicates that the current operator layer has reached an admissible depth limit. Further refinement of the same operator does not lead to improved convergence, implying that progress requires structural extension rather than parameter adjustment.

3 Vacuum Structure and the Cosmological Constant

Gap Class A: Structural Depth

The application of the τ-field framework to vacuum energy produces a suppression of the cosmological constant Λ by approximately ten orders of magnitude relative to naive estimates. This result demonstrates that vacuum energy behaves as an emergent invariant under recursive substrate dynamics.

Despite this substantial suppression, a residual discrepancy of approximately eight orders of magnitude remains. As in the nuclear case, this discrepancy is stable under numerical refinement and parameter variation.

The stability of the vacuum residual indicates that the τ-field captures coarse-grained vacuum structure while leaving finer topological or fluctuation modes unresolved. These modes do not manifest as noise in the projection, but as a systematic offset.

Within UNNS, this behavior is interpreted as evidence that vacuum structure includes internal coherence relations that are not represented at the current operator resolution. The τ-field projects an effective vacuum invariant, but the residual signals unresolved internal degrees of freedom associated with global topology or fluctuation structure.

As in the nuclear case, the appropriate response to this boundary is not further refinement of the existing operator, but extension of the operator

stack to include additional structural modes.

4 Interpretation of Structural Depth

Gap Class A: Structural Depth

The behaviors observed in both nuclear and vacuum regimes exhibit a common signature: convergence of τ-invariants accompanied by stable residual offsets. This pattern indicates that the active operator layer has reached the limit of its admissible resolution while successfully projecting large-scale structure.

Within UNNS, such behavior is not interpreted as incomplete modeling of the same structure, but as evidence that additional internal degrees of freedom exist beneath the resolution of the current operators. These degrees of freedom do not appear as perturbative corrections, but as coherence relations that are invisible to post-projective adjustment.

The distinction between projection failure and depth limitation is therefore crucial. Projection failure would manifest as instability, sensitivity to numerical refinement, or parameter dependence. None of these are observed. Instead, the residuals remain fixed while all other aspects of the dynamics converge.

This stability identifies the residuals as markers of operator depth. They signal that the recursive substrate supports further internal organization that cannot be accessed without extending the operator grammar.

4.1 A.1 Nature of the Missing Modes

The unresolved structure indicated by Gap Class A does not correspond to higher order corrections of existing modes. Rather, it reflects qualitatively distinct internal constraints that govern admissibility prior to τ-projection.

These constraints regulate coherence among internal components of the substrate. Because the τ-field acts only on admissible recursive outputs, any coherence relations that operate before projection remain invisible unless explicitly encoded by an operator acting at that stage.

The missing modes are therefore characterized as pre-projective coherence modes. They do not alter the form of the τ-projection, but restrict the space of structures upon which that projection acts.

4.2 A.2 Operator Coupling and Ordering

The identification of pre-projective coherence modes implies a nontrivial operator ordering. Specifically, any operator regulating internal coherence must act prior to τ-projection.

Formally, this establishes the ordering

$$\kappa \circ \tau \neq \tau \circ \kappa.$$

Collapse and consistency operators, which act on projected invariants, cannot substitute for coherence regulation at the pre-projective stage. Their action presupposes that the projection itself has already occurred.

This ordering distinguishes structural depth effects from post-projective selection or consistency enforcement. It also clarifies why further refinement of collapse criteria does not resolve Gap Class A residuals.

4.3 A.3 Falsifiability of Structural Depth Extensions

The introduction of additional operator layers constitutes a falsifiable extension of the UNNS framework. Structural depth is supported only if the extension modifies the observed residuals without destabilizing previously established invariants.

If the incorporation of pre-projective coherence regulation alters the projection in an uncontrolled manner, destroys convergence of existing τ-invariants, or introduces new instabilities, then the proposed extension must be rejected.

Conversely, if an operator acting prior to τ-projection produces systematic and stable modification of the residuals while preserving the convergence of other invariants, then the interpretation of the residuals as depth boundaries is supported.

This criterion distinguishes genuine structural extension from ad hoc modification. It ensures that additional operators are justified by improved explanatory power rather than by post hoc adjustment.

Quantitative Falsification Criteria. For the purposes of empirical testing, Gap Class A is falsified if no admissible κ-extension satisfies both of the following conditions:

- a reduction of the μ residual by at least 20% relative to its baseline value, and
- preservation of convergence of other established τ-invariants (e.g. α) within $\pm 0.1\%$ of their observed values.

Failure to meet both conditions indicates that the residual is intrinsic to the current τ-field rather than a consequence of missing pre-projective coherence.

Scope Delimitation. Gap Class D, concerning cross-scale coherence and interaction between distinct operator strata, is not addressed in the present work. The analysis here is restricted to single-scale structural depth (Gap Class A) and pre-projective coherence regulation. Cross-scale effects are deferred to future investigation.

5 Minimal κ-Operator Action Schema

The κ-Operator is introduced as a pre-projective structural regulator acting on admissible recursive structures prior to τ-Field projection. Its role is not to generate invariants, but to constrain internal coherence so that subsequent τ-projection operates on structurally admissible inputs.

Let $\mathcal{S}$ denote the space of recursively generable structures admitted by the current UNNS grammar. The κ-Operator acts as an endomorphism

$$\kappa : \mathcal{S} \to \mathcal{S},$$

mapping candidate structures to coherence-regulated structures without introducing new external degrees of freedom.

The κ-Operator does not assign values or invariants. Instead, it restricts admissibility by suppressing or restructuring incompatible internal relations while preserving generability.

By construction, the κ-Operator acts strictly prior to τ-Field projection. The effective projection pipeline is therefore

$$S \xrightarrow{\kappa} S' \xrightarrow{\tau} \tau(S').$$

In general,

$$\tau(\kappa(S)) \neq \kappa(\tau(S)),$$

establishing that coherence regulation and invariant projection are non-commuting operations.

The κ-Operator is assumed to be idempotent on admissible structures,

$$\kappa(\kappa(S)) = \kappa(S),$$

ensuring that coherence regulation converges and does not introduce iterative instability.

6 The κ-Operator as a Coherence Functor

To formalize the role of the κ-Operator independently of any particular instantiation, it is useful to interpret it at the level of abstract structure. In this formulation, the κ-Operator acts as a coherence regulator on the space of admissible recursive structures.

Let **Rec** denote a category whose objects are recursively generable structures admitted by the UNNS grammar, and whose morphisms represent structure-preserving recursive transformations. Let $\mathbf{Rec}_{\text{coh}} \subset \mathbf{Rec}$ denote the full subcategory of internally coherent structures.

The κ-Operator is defined as an idempotent endofunctor

$$\kappa : \mathbf{Rec} \to \mathbf{Rec}_{\text{coh}}.$$

For each object $S \in \mathbf{Rec}$, $\kappa(S)$ is a coherence-regulated representative within the admissibility class of S. For each morphism $f : S \to T$, there exists an induced morphism $\kappa(f) : \kappa(S) \to \kappa(T)$ preserving coherence structure.

Idempotence,

$$\kappa \circ \kappa \cong \kappa,$$

ensures that coherence regulation converges and does not introduce iterative instability.

Let $\tau : \mathbf{Rec}_{\text{coh}} \to \mathbf{Inv}$ denote the τ-Field projection functor mapping coherent structures to emergent invariants. In general,

$$\tau \circ \kappa \neq \kappa \circ \tau,$$

reflecting the non-commutativity of coherence regulation and invariant projection.

7 Operator Grammar Overview

Table 1: UNNS Operator Hierarchy

Operator	**Stage**	**Function**	**Status**
κ	Pre-projective	Coherence regulation	Theoretical
τ	Projective	Invariant projection	Implemented
Collapse	Post-projective	Survival filtering	Implicit
Consistency	Post-projective	Relational enforcement	Partial

8 Toy κ-Instantiation: Graph-Coherence Regulator

To demonstrate the action of the κ-Operator without committing to any physical interpretation, a purely abstract instantiation based on graph coherence is introduced.

Let a recursively generable structure S be represented as a finite weighted graph

$$G(S) = (V, E, w),$$

where vertices represent internal components and weighted edges represent relations between components.

Define a local coherence functional

$$C(G) = \sum_{v \in V} \mathrm{Var}\big(\{w(v,u) \mid (v,u) \in E\}\big),$$

which quantifies inconsistency in relational strength around each vertex.

The κ-Operator acts on $G(S)$ by suppressing incoherent relations according to a fixed admissibility rule. One minimal action is

$$w(v,u) \mapsto \min\big(w(v,u), \overline{w}_v + \delta\big),$$

where $\overline{w}_v$ is the mean weight of edges incident to v and δ is a coherence tolerance parameter.

Edges violating local coherence beyond δ are weakened but not removed, preserving connectivity while enforcing structural regularity.

Repeated application of κ converges,

$$\kappa(\kappa(G)) = \kappa(G),$$

since no further relations violate the coherence bound after regulation.

Let $\tau(G)$ denote any graph-to-invariant projection, such as spectral radius or global connectivity. In general,

$$\tau(\kappa(G)) \neq \tau(G),$$

demonstrating that pre-projective coherence regulation alters emergent invariants without modifying the projection rule.